SANDUNI HULATHDOOWAGE

Colombo, Sri Lanka | skdilukshi@gmail.com | +94 71 772 8723

LinkedIn | GitHub | Google Scholar | Portfolio

EXECUTIVE SUMMARY

Full-Stack Software Engineer with 2 years of professional experience specializing in enterprise application development, business process automation, and scalable web solutions. Experienced in designing, developing, and maintaining complex business applications using modern technologies and frameworks. Skilled in requirements analysis, SRS documentation, database design, UI/UX development, system integration, application security, and CI/CD. Combines practical software engineering experience with research expertise in Artificial Intelligence, Machine Learning, Computer Vision, and OCR, with a strong focus on building reliable, maintainable, and business-oriented software solutions.

WORK EXPERIENCE

Associate Software Engineer | *Upali Group of Companies, Colombo, Sri Lanka* *November 2025 – Present*

- Develop and maintain a full-stack e-commerce platform and enterprise ERP applications supporting multiple business functions.
- Play a key role in the design, development, and implementation of HRIS modules covering employee management, payroll, attendance, leave, loans, promotions, and asset management.
- Implement CRM functionalities, including promotional SMS campaigns, complaint management, and customer relationship workflows.
- Gather stakeholder requirements, analyse business processes, and prepare Software Requirements Specifications (SRS) and functional specifications.
- Design and develop RESTful APIs, database schemas, and integrations with SMS gateways, email services, and third-party systems.
- Implement centralized security mechanisms for managing encryption keys, access tokens, and sensitive data across enterprise applications.
- Develop responsive user interfaces using React.js and backend services using Laravel, optimizing database queries and application performance.
- Contribute to the full Software Development Life Cycle (SDLC), including requirements analysis, development, testing, deployment, and production support, using Git and Agile practices.

Software Engineer Intern | *Web C Lanka, Matugama, Sri Lanka* *July 2025 – October 2025*

- Developed and maintained responsive business websites and software systems, implementing client-specific features, dynamic functionality, and database-driven solutions.
- Developed RESTful APIs and backend modules to support authentication, data management, and third-party service integrations.
- Managed end-to-end deployment and maintenance across hosting and cloud environments, including server configuration, troubleshooting, CI/CD pipeline implementation, and ongoing client support.

Machine Learning Engineer Intern | *Sri Lanka Telecom PLC, Colombo, Sri Lanka* *February 2024 – August 2024*

- Trained and evaluated AI/ML models for the Park & Pay smart parking management system, including vehicle detection, number-plate recognition, and human detection.
- Contributed to Sinhala OCR development using computer vision and deep learning techniques for automated text recognition.
- Contributed to full-stack application development using Laravel and React.js, including backend implementation, database management, and responsive UI/UX design.
- Integrated AI models into full-stack applications and supported Linux-based operations, application deployment, and hosting on DigitalOcean.
- Conducted literature-based technology surveys and reviewed research papers to evaluate and identify optimal technologies, frameworks, and approaches for system development.
- Prepared and delivered weekly project demonstrations to showcase development progress, present implemented features, and gather stakeholder feedback.

EDUCATION

BSc (Hons) in Computing and Information Systems | Second Class Upper Division **FGPA: 3.38 / 4.00 (84.5%)**
Sabaragamuwa University of Sri Lanka October 2020 – April 2025

Achievements: Ranked 1st in the batch in the Fourth Year – First Semester, A+ grade for Undergraduate Internship in Machine Learning, A− grade for Final Year Research (An Enhanced U-Net-based Approach for Sinhala Document Layout Analysis), GPA: 3.72/4.00 in Third Year, GPA: 3.69/4.00 in Fourth Year

---

Professional Certificates

Diploma in Business English (IUHS Campus, Nugegoda, Sri Lanka) Jan 2022 – Jul 2022
Machine Learning for All - University of London (Coursera) 2024
Computer Vision and Image Processing - IBM (Coursera) 2024
Python - University of Moratuwa 2023

Secondary Education

Advanced Level in Physical Sciences Stream Combined Mathematics; Chemistry; Physics
Mahinda Rajapaksha College, Homagama *2018*

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, PHP, JavaScript, TypeScript
- **Frameworks & Libraries:** Laravel, React.js, FastAPI
- **Frontend Development & UI/UX Design:** HTML5, CSS3, Responsive Web Design, Figma
- **Backend Development & Software Engineering:** RESTful API Development, Object-Oriented Programming (OOP), MVC Architecture, Third-Party API Integration
- **Databases & Data Management:** MySQL, MongoDB, Database Design and Management, SQL Query Optimization
- **DevOps, Deployment & Version Control:** Git, GitHub, Docker, Linux, CI/CD, DigitalOcean
- **CMS & Web Optimization:** WordPress, Search Engine Optimization (SEO)

RESEARCH EXPERIENCE

Researcher | Computer Vision for Construction *2025 – Present*

Research Supervisor: Dr. S. Thuseethan and Mr. P. Vignashwaran | Charles Darwin University, Australia

- Investigating computer vision techniques for construction applications, with a focus on cross-site domain generalization and robust object detection under varying visual conditions.
- Exploring domain shifts in construction imagery and image-processing strategies to improve model generalization across construction sites.

Undergraduate Researcher | *Sabaragamuwa University of Sri Lanka* *2025 – Present*

Research Supervisors: Prof. B. T. G. S. Kumara | Sabaragamuwa University of Sri Lanka

- Investigated deep learning-based Sinhala document layout analysis to support region-level document understanding and improve downstream OCR processing.
- Designed and developed an enhanced U-Net architecture integrating Vision Transformer blocks for semantic segmentation of structural elements in Sinhala documents.
- Conducted a systematic literature review on Sinhala document layout and style analysis, synthesizing existing research and contributing to peer-reviewed academic publications.

Research Publications:

1. Hulathdoowage, S., & Kumara, B. (2025). An Enhanced U-Net-based Approach for Sinhala Document Layout Analysis. Journal of Computers and Digital Business, 4(3), 106-114.

2. Hulathdoowage, S., & Kumara, B. (2025). Enhanced U-Net Architecture for Analyzing Sinhala Document Layouts and Styles. ComURS2025 Computing Undergraduate Research Symposium.

3. Hulathdoowage, S., & Kumara, B. (2025). Exploring techniques for analyzing Sinhala document layouts and styles: A systematic literature review. International Journal of Research Publication and Reviews, 6, 7–16.

Research Skills

- **AI & Machine Learning:** Deep Learning, Computer Vision, Machine Learning, Object Detection, Semantic Segmentation, OCR, Predictive Modelling
- **Data Science & Image Processing:** OpenCV, NumPy, Pandas, Matplotlib, Scikit-learn, Image Processing, Data Preprocessing, Data Annotation
- **AI Frameworks & Research Tools:** PyTorch, TensorFlow, Keras, Jupyter Notebook, Google Colab, Kaggle
- **Research Methods:** Systematic Literature Reviews, Research Analysis, Experimental Evaluation, Academic Writing, Research Publication

REFERENCES

Prof. B T G S Kumara
Dean, Faculty of Computing, Sabaragamuwa University of Sri Lanka.
kumara@appsc.sab.ac.lk | +94 71 443 1192

Mr. Chamara Priyadarshana
Senior Software Engineer, Web C Lanka.
chamara@webclanka.com | +94 71 461 9522